Omkar Gulambe

C/C++ Developer | Linux System Engineer | System and Security Programming
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Technical Skills

Languages: C, C++, SQL, Shell Scripting
Frameworks & Libraries: libmagic, libarchive, Poppler, Tesseract OCR, PCRE2, libudev, NFSDK, PAM, fanotify, OpenSSL
OS & Systems: Linux Internals, Kernel Module Development, System Calls, RHEL, Debian, Multithreading, IPC, Thread-Safe Design, epoll, File I/O
Networking: TCP/IP, UDP, Socket Programming, TLS/SNI Parsing, libpcap
Security: Two-Factor Authentication (2FA), PAM, Data Loss Prevention (DLP), Endpoint Security, Device Encryption
Tools: Git, Vim, Make, CMake, GDB, Valgrind, SQLite3
CS Fundamentals: Data Structures and Algorithms, Design Patterns, Object-Oriented Programming

Experience

Software Developer April 2024 – Present
MicroWorld Software Services Pvt Ltd *Andheri, Mumbai*

eScan Antivirus for Linux/Unix (Desktop Edition)

- Involved in designing and developing eScan Antivirus for Linux desktops, a comprehensive security solution to protect Linux machines from viruses and other cyber threats.
- Developed security applications for various Linux distributions (Debian, CentOS, RPM, etc.) using C/C++, focusing on performance and reliability. Built Qt GUI tools in C++ for Linux.

eScan Antivirus for Linux/Unix (Corporate Edition)

- Worked on eScan Antivirus for Linux (Corporate Edition), which collects administrator-defined security policies and product updates from the eScan server to deploy on client endpoints.
- Developed and implemented user-space and kernel-level applications on Linux platforms.

Projects

Attachment Control (Desktop EDR and DLP Edition) | *C/C++, libmagic, libarchive, fanotify*

- Developed an attachment control feature for Linux that monitors and blocks files attached through Chrome, Firefox, SCP, FileZilla, Brave, Thunderbird, Outlook, etc.
- Enforced user-defined policies using file type, extension, and URL-based blocking.

Content Scanner (Desktop DLP Edition) | *C/C++, poppler, tesseract, leptonica, pcre2*

- Developed a data scanning tool that identifies 23 types of confidential data (PAN, Aadhaar, driving license, etc.) across 20+ file types (PDF, images, XML, DOCX, archives, etc.) using regex patterns and OCR.
- Added support to scan password-protected files.
- Blocked Attached files, clipboard activity and printer activity that contain confidential data.

Net filter Development Tool (Desktop EDR and DLP Edition) | *C/C++, NFSDK, Firewall*

- Developed an email control tool for Linux offering global email blocking, recipient-based blocking, and blocking of emails containing attachments. Provided control over email traffic and improve security.
- Provided Website Blocking Functionality.

USB Device Encryption (Corporate DLP Edition) | *C/C++, Linux Kernel, libudev, BIO, VFS*

- Developed a Linux daemon that automatically encrypts and decrypts USB storage devices, enforcing corporate-only access by preventing encrypted USB drives from being accessed on unauthorized systems.
- Also Implemented kernel-level encryption and decryption of USB storage I/O by intercepting the BIO and VFS layers, transparent to existing applications and independent of the file system.

Linux Two-Factor Authentication (2FA) | *C++, PAM*

- Implemented PAM-based OTP authentication for SSH and local Linux login.
- Supported per-user and system-wide policy configurations.
- Integrated time-based OTP verification without disrupting existing authentication workflows.

Education

D.Y. Patil College of Engineering & Technology, Kolhapur 2020 – 2024
B.Tech, CGPA: 7.96